MILITARY SPECIFICATION

DRIVE, BALLISTICS, XM15

1. SCOPE

1.1 This specification covers type of instrument known as the Drive, Ballistics, XM15.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-F-13926 Fire control Materiel; General Specification Governing. the Manufacture and Inspection of

MIL-P-14232 –Parts, Equipment and Tools for Ordnance Materiel; Packaging of

MIL-I~45208 –Inspection Requirements, General Specification for

STANDARDS

MILITARY

MIL-STD-105 –Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-109 –Quality Assurance Terms and Definitions

DRAWINGS

U. S. ARMY MUNITIONS COMMAND

F10616375 –Drive, Ballistics, XM15

LSQAP 10516375 –List of Supplementary Quality Assurance Provisions for Drive, Ballistics, XM15

IEL 10516375 –List of Inspection Equipment Numbers for Drive, Ballistics, XM15

PACKAGING DATA SHEET

MIL-P-14232/P1051635--Packaging of Drive, Ballistics, X15

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENT'S

3.1 Materials. Materials shall be in accordance with drawings, material specifications and general specifications forming a part of this specification.

3.2 Fabrication. The ballistics drive shall be manufactured in accordance with Drawing F10516375, "Drive, Ballsties, XM15", and drawings pertaining thereto. The requirements of this specification are detailed only to the extent considered necessary to obtain the desired mechanical characteristics, performance and permanence to insure inherent quality and performance capabilities.

3.3 General Specification. The contractor shall be responsible for adherence to, and compliance with, the following requirements of MIL-F-13926:

(a) Order of precedence.
(b) Dimensions and tolerances.
(c) Inorganic protective surface finishes.
(d) Part identification and marking.
(e) Workmanship.

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3.4 Envirmental.

3.4.1 Operating Tenperature. The ballistics drive shall meet the requirements of 3.6.2 ,while exposed to and thermally stabilized at ambient temperatures of --40° and +150 ° Fahrenheit.

3.5 Vibration. The ballistics drive shall meet the requirement of 3.6 after being vibrated for five minutes plus or minus 15 Seconds at an amplitude. of 1/16 inch (3/8 inch total excursion) at a frequency of 30 Cycles per second.

3.6 Performance. With the ballistics drive mounted in the initial operating position shown on Drawing F10516375 and with the double eccentric -adjusted to the drawing requirement, the output coupling and Arm 1051-6274 positioned as shown on Drawing D10516381; any angular movement of a simiulated gum trunnion eleven inch linkage shall a ,uate the drive throvgh the following ranges:

(a) Clockwise movement of the simulated Iinkage to 17° minimum and return “to initial position.

(b) Counterclockwise movement Of the simulated linkage to 5° and return to initial position,

3.6.1 Backlash at any position within the 22° range shall not exceed ‘0.1 roil.

3.6.2 Operability. The torque for all rotating. studs in the ballistics drive linkage throughout the operating ranges specified shall not exceed 5 inch pounds plus or minus 0.5 inch pounds when exposed and thermally stabilized at ambient temperatures of —400 and +150° Fahrenheit.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or order, the supplier is responsible for the performante of-a]] inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own any other inspection facilities and services, acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirement

4.1.1 General provisions. The quality assurance provisions of this specification and of other documents referenced herein form. the basis for inspection to be performed by the supplier. Inspection shall be in accordance with the requirements of MIL-I-145208 and the component inspection requirements of MIL-F-13926. Definitions of inspection terms not otherwise defined herein shall be .as listed in MIL-STD-109.

4.2 Inspection provisions.

4.2.1 Unless otherwise specified by the contracting officer, inspection lot size, formation, and presentation of lots shall be in accordance with “Submission of Product” as specified in MIL-STD-105.

4.2.2 Components and subassemblies shall be inspected in accordance with the inspection provisions contained in MIL-F-13926 and List of Supplementary Quality Assurance Provisions, LSQAP 10516375. Examinations and tests related to Section 3 herein shall be performed on a defect (individual characteristic) basis in accordance with MIL-STD-105, and the inspection level and sampling plans specified in table 1 titled “Classification of Defects”. Examinations and test for Packaging, Packing and Marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier prior to Government acceptance or rejection of lot. The Government reserves the right to inspect for any applicable requirement, and to reject individual non-conforming items.

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TABLE I. *Classification of Defects*

Use Level II of table 1 with Sampling Plan Table II-A of MIL-STD-106.

	Requirement	Test procedure
CRITICAL: NONE		
MAJOR: AQL 1.0 Per cent Defective		
101. Vibration	3.5	4.5.1
102. Performance	3.6	4.5.2
103. Backlash	3.6.1	4.5.3
104. Torque (Standard Ambient)	3.6.2	4.5.4

4.2.3 Disposition of non-conf orming product. Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.3 Control tests. Three ballistics drives shall be selected at random from each 100 produced, or from each month's production, whichever occurs first. Control samples will have successfully met all acceptance tests specified herein prior to conducting the following examinations and tests:

Control test	Requirement	Test Procedure
1. Materials	3.1	As per specification or drawing
2. Fabrication	3.2	Visual-per drawings
3. General Specification	3.S	Visual-MIL-F-13926
4. Operating Temperatures	3.4.1	4.6.5

4.3.1 Control test failure. Should the control test sample fail to meet the specified test requirements, acceptance of the product will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved.

4.4 Inspection equipment. Except as otherwise provided for by the contract, the contractor shall furnish and maintain all required measuring and testing equipment in accordance with the "Inspection Equipment" requirement of MIL-A-45208. The Government reserves the right to use the test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractor's delivery schedule.

4.4.1 Contractor furnished inspection equipment.

4.4.1.1 Government design. All inspection equipment required by drawings forming a part of the contract and not provided by the Government shall be supplied by the contractor in accordance with the list of Inspection Equipment Numbers, IEL 10516375. Alternate designs may be substituted only as provided by "Inspection Provisions" of MIL-I-45208.

4.4.1.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the"Test Methods and Procedures" specified in 4.5, and with the "Component Inspection" and "Test Facilities" requirements of MIL-F-13926. Concept, construction, materials, dimensions, and tolerances used in design of test equipment shall be so selected and controlled as to insure that the test equipment will permit positive rejection of a product which exceeds the prescribed tolerance limit, and will reliably indicate acceptability of a product which does not exceed 90 per cent of the prescribed tolerance limit. Construction shall be such as to facilitate routine calibration of the equipment.

4.5 Test methods and procedures.

4.5.1 Vibration. This test shall be conducted with the ballistics drive mounted in a suitable holding fixture in the initial operating position as shown on Drawing FI0516375. The drive shall be vibrated as specified in 3.5 and at the completion of the vibration, the drive shall be subjected to a visual-tactile examination for damage. The drive shall then be subjected to tests in 4.5.2 to 4.5.4 inclusive.

4.5.2 Performance. To determine conformante to 3.6, the drive shall be mounted in a fixture, the accuracy of which shall be compatible to the accuracy shown on fixture Drawing F10515780, with the arms and linkages positioned as specified on the fixture drawing. The fixture shall contain an 11.000 inch simulated gun trunnion linkage arm pivoting about a simulated gun trunnion.

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With this simulated linkage attached to the drive and from the initial operating position of the output coupling, the following shall be performed:

(a) Rotate simulated linkage in a clockwise direction a minimum of 17 degrees and return as specified in 3.6.

(b) Rotate simulated linkage in a counterclockwise direction a minimum of 5 degrees and return as specified in 3.6.

The drive shall meet the clockwise (elevation) and counterclockwise (depression) requirements of 3.6 when operated and measured from the initial operation position.

4.5.3 Baokksh. To determine conformance to 3.6.1 this check shall be accomplished at the initial operating position, the 17° clockwise point and the 5° counterclockwise point. The backlash in the drive shall not exceed the specified requirement of 3.6.1. This test may be accomplished in conjunction with 4.5.2.

4.5.4 Torque. To determine conformance to 8.6.2, attach stmdard torque measuring instruments and appropriate adapters to each rotating stud. Actuate the drive through the range specified in 3.6. This test. shall be performed at standard ambient conditions when testing in accordance with 4.2.2, table I except when 4.3 and 4.6.5 is applicable,

4.5.5 Operating temperature, The test equipment shall be in accordance with the "Test Facilities" requirements of MIL-F 13926. The temperatures shall be in accordance with the requirements of 3.4.1. When the drive is thermally stabilized at each indicated temperature, it shall be subjected to the tests in 4.5.2 to 4.5.4 inclusive to determine conformance to 3.4.1.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. Packaging, packing and marking shall be in accordance with MIL-P-14232 and Packaging Data Sheet MIL-P-14232/P10516375. The level. of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. The ballistics drive covered herein acts as a solid linkage providing angular movement of the gunner's periscope corresponding to the movement of the gun in elevation or depression. This drive is primarily intended for use in the T118EI combat engineer's vehicle.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) Selection of applicable levels of preservation, packaging and packing.

(c) Applicable Stock number,

Custodian:
Army-MU

Preparing activity:
Army-MU
Project No. 1220-A152Y

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 15419

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